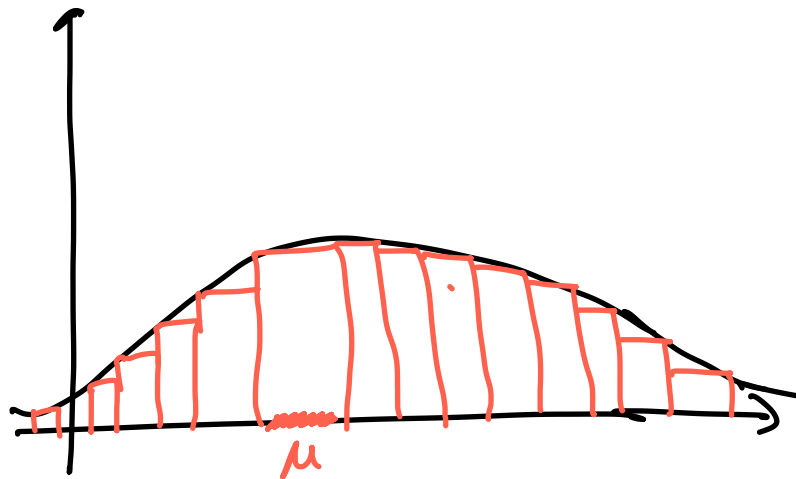


The Lebesgue Integral on $\mathbb{R}^n$

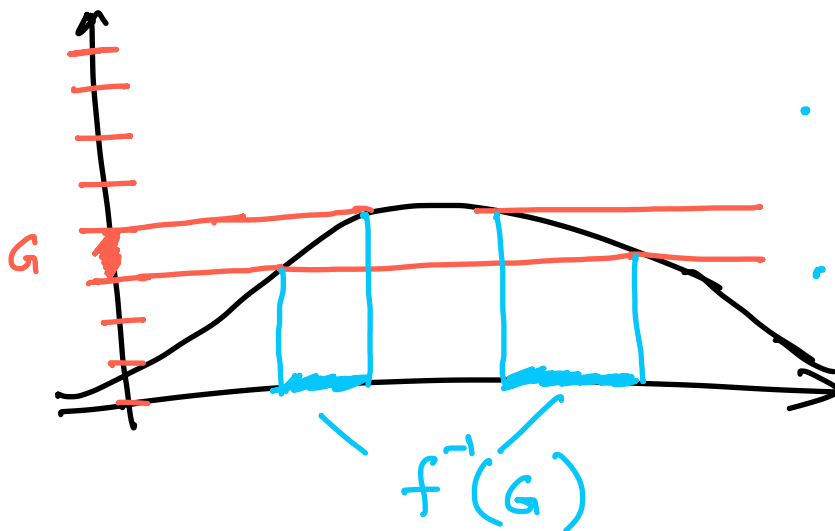
Intuition:

Riemann



- bounded
- continuous
- finite set of rectangles.

Lebesgue

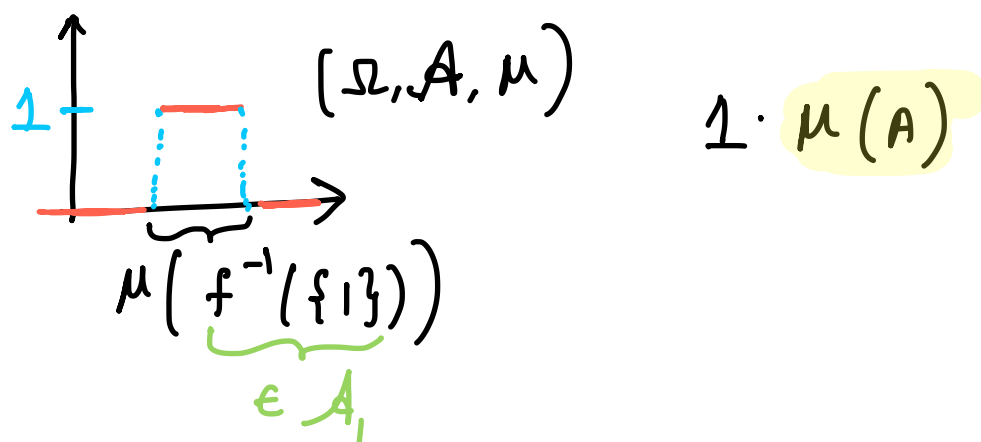


- not bounded
- need not be continuous
- countable sets

Def: A function $f: (\Omega_1, \mathcal{A}_1) \rightarrow (\Omega_2, \mathcal{A}_2)$ between two measurable spaces is called a measurable if pre-images of measurable sets are measurable:

$$\forall A_2 \in \mathcal{A}_2 : f^{-1}(A_2) \in \mathcal{A}_1$$

$$\hookrightarrow =: \{x \in \Omega_1 \mid f(x) \in A_2\}$$



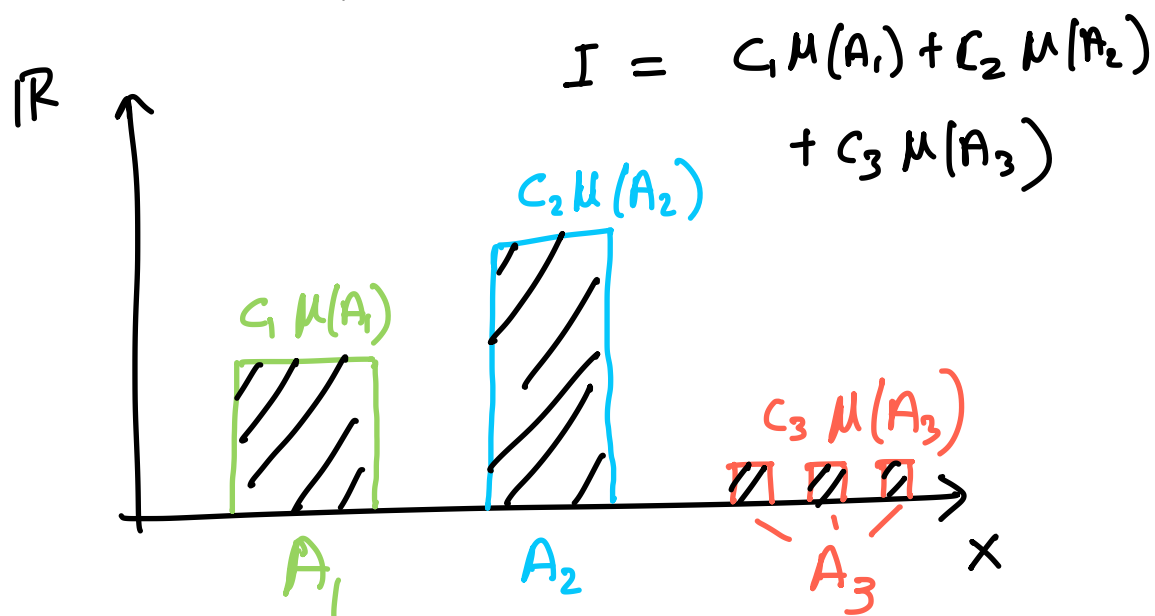
$$(\Omega, \mathcal{A}), (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

Characteristic function (also indicator func.)

$$\chi_A: \Omega \rightarrow \mathbb{R}, \chi_A(\omega) := \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

Def: $\phi: \mathbb{R}^n \rightarrow \mathbb{R}$ is called a simple function if there exist measurable sets $A_i \subset \mathbb{R}^n$, $c_i \in \mathbb{R}$ such that

$$\phi(x) = \sum_{i=1}^n c_i \chi_{A_i}(x)$$



$$\phi(x) = c_1 \chi_{A_1}(x) + c_2 \chi_{A_2}(x) + c_3 \chi_{A_3}(x)$$

$$I(\phi) = \int \phi d\mu = \sum_{i=1}^n c_i \mu(A_i) \rightarrow \text{Lebesgue integral for simple func.}$$

Problem: $3 \cdot \infty - 2 \cdot \infty$??

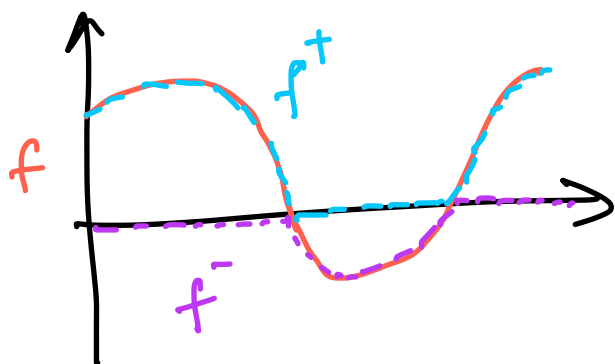
For a function $f^+ : \mathbb{R}^n \rightarrow [0, \infty)$ we define its Lebesgue integral

$$\int f^+ d\mu = \sup \left\{ \int \phi d\mu \mid \phi \leq f, \phi \text{ simple} \right\}$$

(might be ∞)

For a general function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ we split the function into positive and negative parts: $f = f^+ - f^-$, $f^+ \geq 0$, $f^- \geq 0$

$$\text{where } f^+(x) = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ 0 & \text{otherwise} \end{cases}$$



$$f = f^+ - f^-$$

Note: f^+ , f^- are measurable if f is measurable.

If both f^+ and f^- satisfy $\int f^+ d\mu < \infty$
 $\int f^- d\mu < \infty$, then we call f integrable
 and define

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu$$

Much more powerful notion than Riemann Integral.

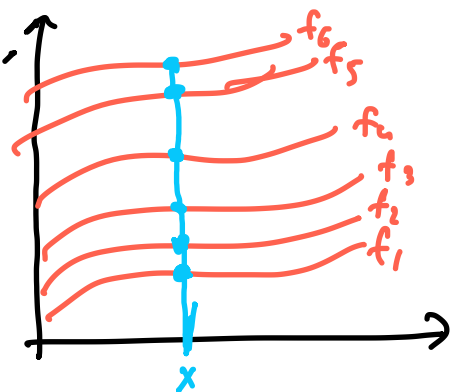
Example: $\int \chi_{\mathbb{Q}} d\mu = 1 \cdot \mu(\mathbb{Q}) = 0$

Two important Theorems

Theorem (monotone convergence): Consider a
 sequence of functions $f_n : \mathbb{R}^n \rightarrow [0, \infty)$ that is
 pointwise non-decreasing: $\forall x \in \mathbb{R}^n, f_{k+1}(x) \geq f_k(x)$.

Assume that all f_k are measurable,
 and that the pointwise limit exists

$$\forall x : \lim f_k(x) =: f(x)$$



Then:

$$\int \lim_{k \rightarrow \infty} f_k(x) dx = \lim_{k \rightarrow \infty} \int f_k(x) dx$$

$\Downarrow$

$$\int f(x) dx$$

Theorem (dominated convergence):

$f_k : B \rightarrow \mathbb{R}$, $|f_k(x)| \leq g(x)$ on B , $g(x)$ is integrable. Assume that the pointwise limit exists: $\forall x \in B$, $f(x) := \lim_{n \rightarrow \infty} f_n(x)$.

Then:

$$\int \lim_{k \rightarrow \infty} f_k(x) dx = \lim_{k \rightarrow \infty} \int f_k(x) dx$$

$\Downarrow$

$$\int f(x) dx$$

Partial Derivatives on $\mathbb{R}^n$

Consider $f: \mathbb{R}^n \rightarrow \mathbb{R}$

Def: f is called partially differentiable with respect to variable x_j at point $\xi \in \mathbb{R}^n$ if the function

$$x_j \mapsto g(x_j) := f(\xi_1, \xi_2, \dots, \xi_{j-1}, x_j, \xi_{j+1}, \dots, \xi_n)$$

$$g: \mathbb{R} \rightarrow \mathbb{R}$$

is differentiable at $\xi_j \in \mathbb{R}$. j-th unit vector = $\begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \rightarrow j$

Notation:
$$\frac{\partial f(\xi)}{\partial x_j} = \lim_{h \rightarrow 0} \frac{f(\xi + e_j \cdot h) - f(\xi)}{h}$$
 ↑ $\in \mathbb{R}$

$$\mathbb{R}^n \Rightarrow x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad f(x) = x_1^2 + x_2^2 \cdot x_1$$

$f: \mathbb{R}^2 \rightarrow \mathbb{R}$

If all partial derivatives exist, then the vector of all partial derivatives is called the gradient:

$$\text{grad}(f)(\xi) = \nabla f(\xi) = \begin{pmatrix} \frac{\partial f(\xi)}{\partial x_1} \\ \vdots \\ \frac{\partial f(\xi)}{\partial x_n} \end{pmatrix} \in \mathbb{R}^n$$

If $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, we decompose f into its m component functions $f = \begin{pmatrix} f_1 \\ \vdots \\ f_m \end{pmatrix}$. We

define the Jacobian matrix:

$$Df(x) = \begin{pmatrix} \boxed{\frac{\partial f_1}{\partial x_1} \quad \dots \quad \frac{\partial f_1}{\partial x_n}} \\ \vdots \\ \boxed{\frac{\partial f_m}{\partial x_1} \quad \dots \quad \frac{\partial f_m}{\partial x_n}} \end{pmatrix} \in \mathbb{R}^{m \times n}$$

$\nabla f_1(x)$
 $\nabla f_m(x)$

 Even if all partial derivatives exist at ξ , we do not know if f is continuous at ξ .

Example: $f: \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$f(x, y) = \begin{cases} \frac{x \cdot y}{x^2 + y^2}, & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } x = y = 0 \end{cases}$$

For $(x, y) \neq (0, 0)$

$$\nabla f(x, y) = \left(y \cdot \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad x \cdot \frac{x^2 - y^2}{(x^2 + y^2)^2} \right)$$

$\nabla f(0, 0) = 0$ since $f(x, 0) = 0 \quad \forall x$, $f(0, y) = 0 \quad \forall y$
but f is not continuous at 0.

Total Derivative

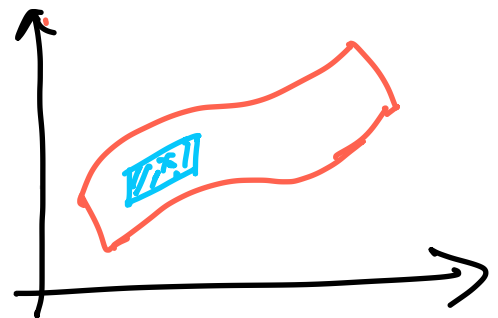
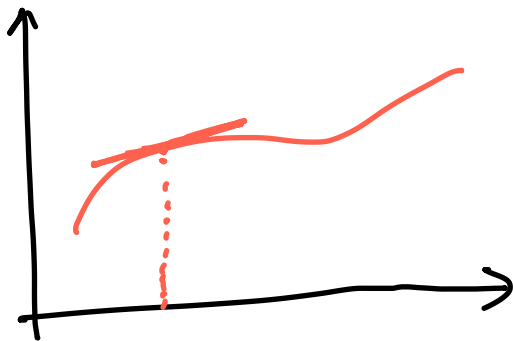
$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad \xi \in U$$

f is differentiable at ξ if there exists a linear mapping $L: \mathbb{R}^n \rightarrow \mathbb{R}^m$ such that

$$\text{for } h \in \mathbb{R}^n,$$

$$f(\xi + h) - f(\xi) = L(h) + r(h)$$

$$\text{with } \lim_{h \rightarrow 0} \frac{r(h)}{|h|} \rightarrow 0.$$



Intuition: f is "locally linear"

Theorem: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at ξ

- Then f is continuous at ξ
- The linear functional L coincides with the gradient:

$$\begin{aligned} f(\xi + h) - f(\xi) &= \sum_{j=1}^n \frac{\partial f}{\partial x_j}(\xi) \cdot h_j + o(h) \\ &= \langle \nabla f(\xi), h \rangle + o(h) \end{aligned}$$

If $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, it is differentiable if all coordinate functions $f_1, f_2, \dots, f_m$ are differentiable. Then all partial derivatives exist and $L(h) = (\text{Jacobian matrix}) \cdot h$

Theorem: If all partial derivatives exist and are all continuous, then f is differentiable.



If partial derivatives exist, but are not continuous, then f doesn't need to be differentiable.

Directional Derivatives

Def: Assume $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is continuously differentiable, $\mathbf{v} \in \mathbb{R}^n$ with $\|\mathbf{v}\| = 1$. The directional derivative of f at ξ in the direction of $\mathbf{v}$ is defined as,

$$D_{\mathbf{v}} f(\xi) = \lim_{t \rightarrow 0} \frac{f(\xi + \overset{\substack{\in \mathbb{R} \\ \downarrow}}{t} \cdot \overset{\substack{\in \mathbb{R}^n, \text{ direction} \\ \downarrow}}{\mathbf{v}}) - f(\xi)}{t}$$

Theorem: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ differentiable at ξ .

Then all the directional derivatives exist, and we can compute them as:

$$D_{\mathbf{v}} f(\xi) = \left(\nabla f(\xi) \right)^T \cdot \mathbf{v} = \sum_{i=1}^n \overset{\substack{\in \mathbb{R} \\ \downarrow}}{v_i} \overset{\substack{\text{partial} \\ \text{derivative.}}}{\frac{\partial f}{\partial x_i}(\xi)}$$

$\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$

The largest value of all directional derivatives is

attained in direction: $\mathbf{v} = \frac{\nabla f(\xi)}{\|\nabla f(\xi)\|}$